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Chapter 24

Portfolio Performance Evaluation

INVESTMENTS THIRTEENTH
EDITION

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Overview

- If markets are efficient, investors must be able to measure performance of their asset managers.
- Discuss methods to evaluate investment performance.
- Conventional approaches to risk adjustment.

Time-Weighted Returns

- **Time-weighted average**
 - Geometric average is a time-weighted average.
 - Each period's return has equal weight.

$$(1 + r_G)^n = (1 + r_1) \times (1 + r_2) \times \dots \times (1 + r_n)$$

→

$$r_G = \left[(1 + r_1) \times (1 + r_2) \times \dots \times (1 + r_n) \right]^{1/n} - 1$$

Dollar-Weighted Returns ¹

- **Dollar-weighted rate of return** is the internal rate of return on an investment.
- Returns are weighted by the amount invested in each period.

$$PV = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots \frac{C_n}{(1+r)^n}$$

Time-Weighted and Dollar-Weighted Returns: Example ¹

A stock sells for \$50. You purchase one share today and one more in one year. At the end of the second year, you sell both shares for \$54. Dividends of \$2 per share are paid annually at the end of each year (but before shares are sold).

Time	Outlay
0	\$50 to purchase first share
1	\$53 to purchase second share a year later
Proceeds	
1	\$2 dividend from initially purchased share
2	\$4 dividend from the 2 shares held in the second year, plus \$108 received from selling both shares at \$54 each

Time-Weighted and Dollar-Weighted Returns: Example ₂

The **time-weighted** (geometric average) return is 7.81%:

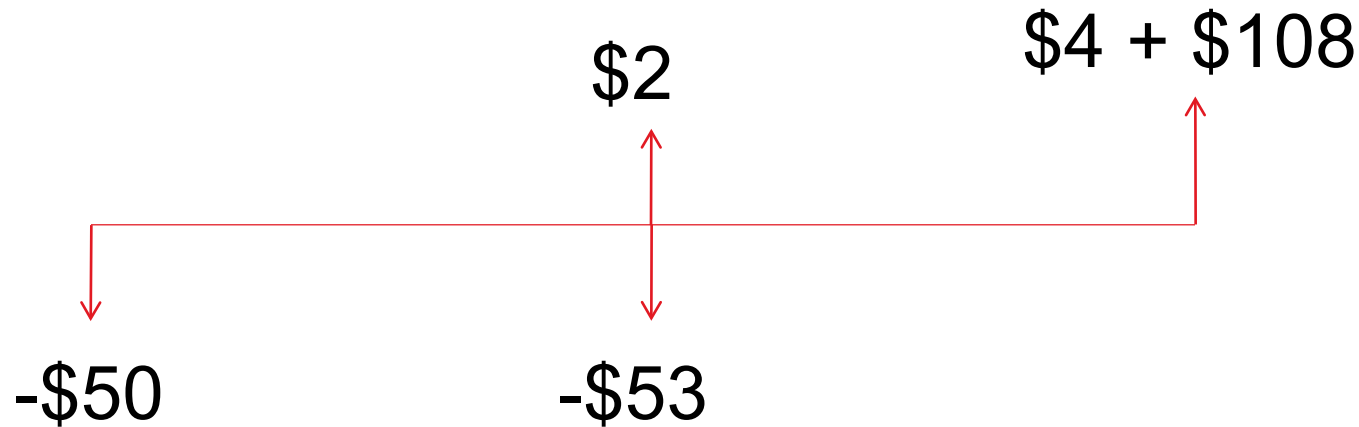
$$r_1 = \frac{P_1 - P_0 + D_1}{P_0} = \frac{\$53 + 2 - 50}{\$50} = .1000$$

$$r_2 = \frac{P_2 - P_1 + D_2}{P_1} = \frac{\$54 + 2 - 53}{\$53} = 0.0566$$

$$r_{\text{Time-Weighted}} = [(1 + r_1) \times (1 + r_2)]^{.5} - 1 = [1.10 \times 1.0566]^{.5} - 1 = .0781$$

*Note that arithmetic average can also be used to compute time-weighted return

Time-Weighted and Dollar-Weighted Returns: Example ³



The **dollar-weighted** (IRR) return is 7.12%:

$$\$50 + \frac{\$53}{1 + r_{\text{DollarWeighted}}} = \frac{\$2}{1 + r_{\text{DollarWeighted}}} + \frac{2 \times (\$54 + \$2)}{(1 + r_{\text{DollarWeighted}})^2}$$

$$r_{\text{DollarWeighted}} = .0712$$

The dollar-weighted average is less than the time-weighted average in this example because more money is invested in year two, when the return was lower

Adjusting Returns for Risk

- Simplest and most popular way to adjust for risk is to compare rates of return with those of other investment funds with similar risk characteristics.
- **Comparison universe** is the set of money managers employing similar investment styles, used for assessing the relative performance of a portfolio manager. (e.g. high-yield bond funds)

Figure 24.1 Universe Comparison

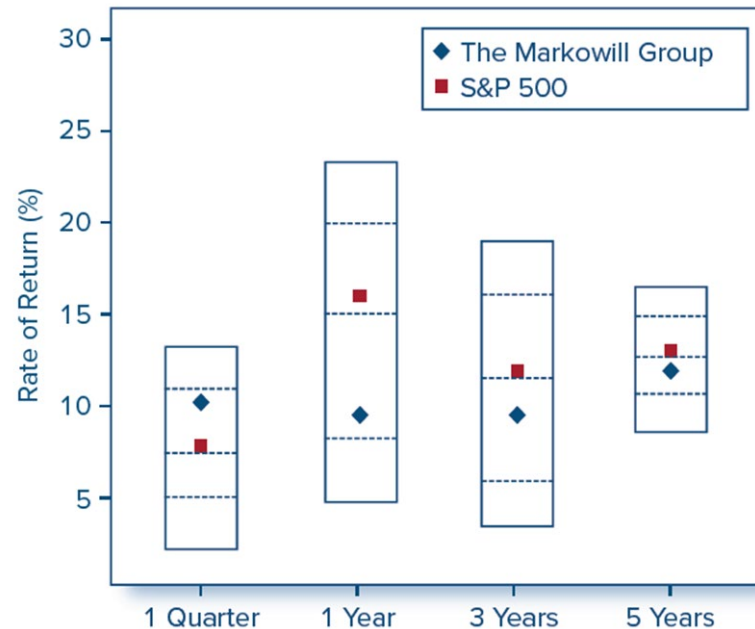


Figure 24.1 Universe comparison, periods ending December 31, 2028

Risk-Adjusted Performance: Sharpe

- **Sharpe's ratio** divides average portfolio excess return over the sample period by the standard deviation of returns over that period.
- Measures reward to (total) volatility trade-off:

$$S_P = \frac{(\bar{r}_P - \bar{r}_f)}{\sigma_P}$$

Risk-Adjusted Performance: Treynor

- **Treynor's measure** is a ratio of excess return to beta, like the Sharpe ratio, but it uses systematic risk instead of total risk.

$$T_P = \frac{(\bar{r}_P - \bar{r}_f)}{\beta_P}$$

Risk-Adjusted Performance: Jensen

- **Jensen's alpha** is the average return on the portfolio over and above that predicted by the CAPM, given the portfolio's beta and the average market return.

$$\alpha_P = \bar{r}_P - \left[\bar{r}_f + \beta_P (\bar{r}_M - \bar{r}_f) \right]$$

- **Jensen's alpha** is the average return on the portfolio over and above that predicted by the Fama-French three-factor model, given the portfolio's beta and the average market return.

$$\alpha_P = \bar{r}_P - \left(\bar{r}_f + \beta_M (\bar{r}_M - \bar{r}_f) + \beta_{SMB} \overline{SMB} + \beta_{HML} \overline{HML} \right)$$

Risk-Adjusted Performance: Information Ratio

- **Information ratio** divides the alpha of the portfolio by the nonsystematic risk of the portfolio, called “tracking error” in the industry.
- Measures abnormal return per unit of risk that in principle could be diversified away by holding a market index portfolio.

$$\text{Information Ratio} = \frac{\alpha_P}{\sigma(e_P)}$$

$$S_P^2 = S_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

M Squared Measure and the Shape Ratio

- Focuses on total volatility as a measure of risk, but its risk adjustment leads to an easy-to-interpret differential return relative to the benchmark index.

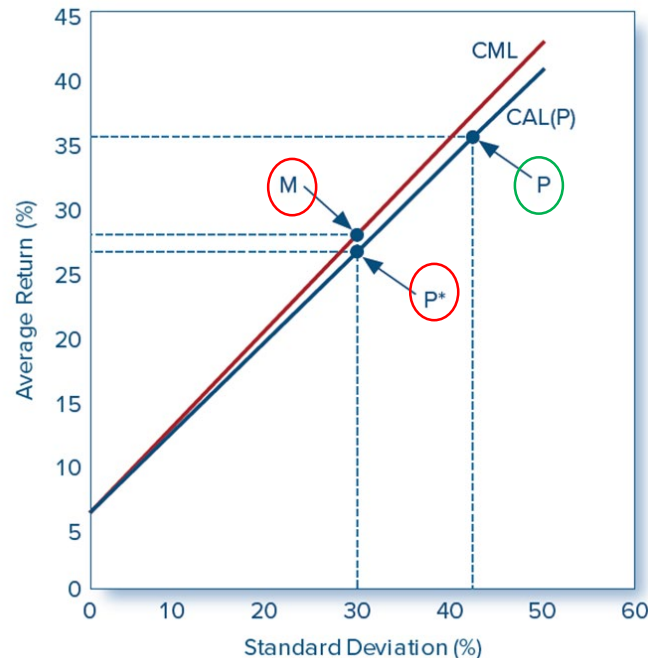
$$M_P^2 = r_{P^*} - r_M$$

Where:

P^* = Portfolio adjusted to σ_M

r_{P^*} = Return of adjusted portfolio P^*

Figure 24.2 The M Squared of Portfolio P



$$\frac{\overline{(r_P - r_f)}}{\sigma_P} \quad \text{Sharpe ratio : Slope}$$

Create an adjusted portfolio P^* that combines P with Treasury Bills

Set P^* to have the same standard deviation as the market index

Now compare market and P^* returns:

$$M_P^2 = r_{P^*} - r_M$$

Figure 24.2 The M^2 of Portfolio P is negative even though its average return was greater than that of the market index, M .

M^2 Measure: Example

Managed Portfolio P: $r_P = 35\%$ $\sigma_P = 42\%$

Market Portfolio: $r_M = 28\%$ $\sigma_M = 30\%$

T-bill return = 6%

P* Portfolio: $30/42 = .714$ in P and $.286$ in T-bills

$$r_{P^*} = (.714) \times (.35) + (.286) \times (.06) = 26.7\%$$

$$M^2 = r_{P^*} - r_M = 26.7\% - 28\% = -1.3\%$$

$r_{P^*} < r_M \rightarrow$ the managed portfolio underperformed

T² (Treynor Square) Measure

- Used to convert the Treynor Measure into percentage return basis
- Makes it easier to interpret and compare
- Equates the beta of the managed portfolio with the market's beta of 1 by creating a hypothetical portfolio made up of T-bills and the managed portfolio
- If the beta is lower than one, leverage is used and the hypothetical portfolio is compared to the market

T² Example

	Port. P.	Market
Risk Prem. ($r-r_f$)	13.00%	10.00%
Beta	0.80	1.0
Alpha	5.00%	0.00%
Treynor Measure	16.25	10.00

Weight to match Market $w = \beta_M / \beta_P = 1.0 / 0.8$

Adjusted Return $R_P^* = w (R_P) = 16.25\%$

$T_P^2 = R_P^* - R_M = 16.25\% - 10\% = 6.25\%$

Appropriate Performance Measure

It depends on investment assumptions

1) If P is not diversified, then use the Sharpe measure as it measures reward to risk (M^2)

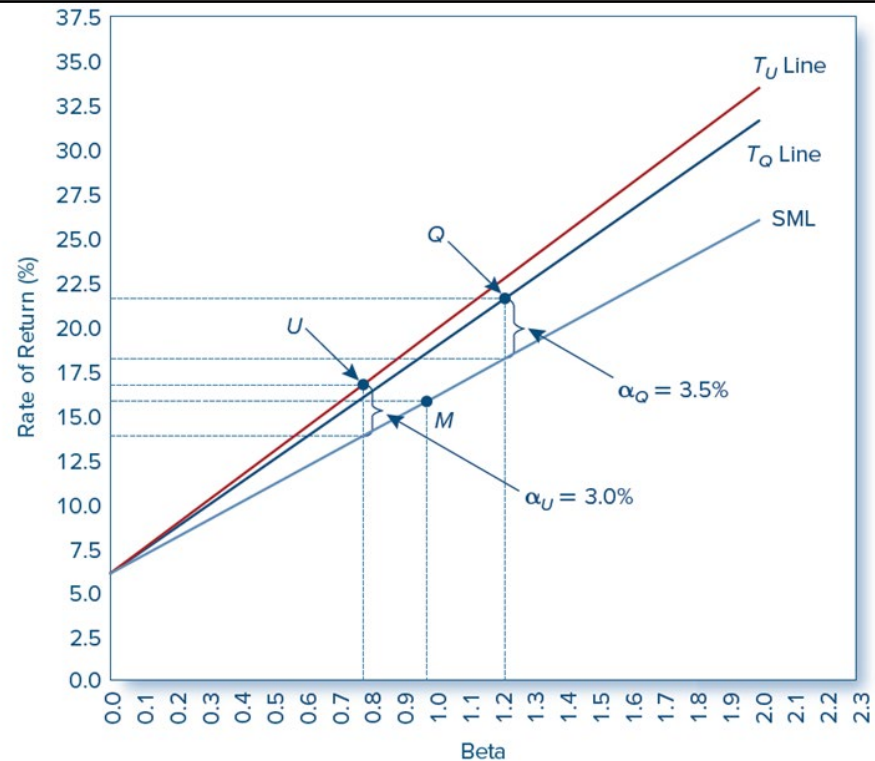
2) If the P is diversified, nonsystematic risk is negligible and the appropriate metric is Treynor's, measuring excess return to beta (T^2 , Jensen's α)

Performance Measure	Definition	Application
Sharpe	$\frac{\text{Excess return}}{\text{Standard deviation}}$	When choosing among portfolios competing for the overall risky portfolio
Treynor	$\frac{\text{Excess return}}{\text{Beta}}$	When ranking many portfolios that will be mixed to form the overall risky portfolio
Information ratio	$\frac{\text{Alpha}}{\text{Residual standard deviation}}$	When evaluating a portfolio to be mixed with the benchmark portfolio

Figure 24.3 Treynor Measures of Two Portfolio and the Market Index

	Risk-free Asset	Portfolio Q	Portfolio U	Market Index, <i>M</i>
Beta	0	1.3	0.8	1.0
Average return	6	22.0	17.0	16.0
Excess return (%)	0	16.0	11.0	10.0
Alpha (%)	0	3.5	3.0	0.0

Is Q better than U?



The Role of Alpha in Performance Measures

Alpha and the Sharpe measure

ρ = Correlation between R_P and R_M

$$S_P = \frac{\beta_P \bar{R}_M}{\sigma_P} + \frac{\alpha_P}{\sigma_P} \quad \beta_P = \frac{\rho \sigma_P \sigma_M}{\sigma_M^2} = \rho \frac{\sigma_P}{\sigma_M}$$

$$S_P - S_M = \rho \frac{\sigma_P}{\sigma_M} \frac{\bar{R}_M}{\sigma_P} + \frac{\alpha_P}{\sigma_P} - S_M = S_M (\rho - 1) + \frac{\alpha_P}{\sigma_P}$$

- Positive alpha does not guarantee a higher Sharpe than the benchmark because $S_M(\rho-1) < 0$.
- Necessary but not a sufficient condition for net performance improvement
 - The alpha must be large enough to offset increase in residual risk from moving away from the diversified optimum.

The Role of Alpha in Performance Measures

Alpha and the Treynor measure

$$T_P = \frac{\bar{R}_P}{\beta_P} = \frac{\beta_P \bar{R}_M + \alpha_P}{\beta_P} = \bar{R}_M + \frac{\alpha_P}{\beta_P} \quad \beta_M = 1 \quad T_M = \bar{R}_M$$

$$T_P - T_M = \frac{\alpha_P}{\beta_P}$$

- Positive alpha does not guarantee a higher Treynor ranking because you have to know the Beta as well.

The Role of Alpha in Performance Measures

- A positive alpha is necessary to outperform the passive market index
 - Though necessary, it's not enough to guarantee a portfolio will outperform the index.

Most widely used performance measure

	Treynor (T_p)	Sharpe* (S_p)	Information Ratio
Relation to alpha	$\frac{E(r_p) - r_f}{\beta_p} = \frac{\alpha_p}{\beta_p} + T_M$	$\frac{E(r_p) - r_f}{\sigma_p} = \frac{\alpha_p}{\sigma_p} + \rho S_M$	$\frac{\alpha_p}{\sigma(e_p)}$
Improvement compared to market index	$T_p - T_M = \frac{\alpha_p}{\beta_p}$	$S_p - S_M = \frac{\alpha_p}{\sigma_p} - (1 - \rho)S_M$	$\frac{\alpha_p}{\sigma(e_p)}$

* ρ denotes the correlation coefficient between portfolio P and the market and is less than 1.

Table 24.2 Excess Returns for Portfolios P and Q and the Benchmark M over 12 Months

Performance Statistics

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Month	Portfolio <i>P</i>	Alternative <i>Q</i>	Index <i>M</i>
1	3.58%	2.81%	2.20%
2	-4.91	-1.15	-8.41
3	6.51	2.53	3.27
4	11.13	37.09	14.41
5	8.78	12.88	7.71
6	9.38	39.08	14.36
7	-3.66	-8.84	-6.15
8	5.56	0.83	2.74
9	-7.72	0.85	-15.27
10	7.76	12.09	6.49
11	-4.01	-5.68	-3.13
12	0.78	-1.77	1.41
Average	2.77	7.56	1.64
Standard deviation	6.45	15.55	8.84

Table 24.2

Excess returns for portfolios *P* and *Q* and the market index *M* over 12 months

Table 24.3 Performance Statistics: Example

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	Portfolio <i>P</i>	Portfolio <i>Q</i>	Portfolio <i>M</i>
Sharpe ratio	0.43	0.49	0.19
M^2	2.16	2.66	0.00
SCL regression statistics			
Alpha	1.63	5.26	0.00
Beta	0.70	1.40	1.00
Treynor	3.97	5.38	1.64
T^2	2.34	3.74	0.00
$\sigma(e)$	2.02	9.81	0.00
Information ratio	0.81	0.54	0.00
<i>R</i> -square	0.91	0.64	1.00

Table 24.3

Performance statistics

Interpretation of Performance Statistics

- If P or Q represents the entire investment, Q is better because of its higher Sharpe measure and better M^2 .
- If P and Q are potential sub-portfolios, Q also dominates because its Treynor measure is higher.
- If we seek an active portfolio to mix with an index portfolio, P is better due to its higher information ratio.

Realized Returns Versus Expected Returns

- Must determine “significance level” of a performance measure to know whether it reliably indicates ability
 - To estimate the portfolio alpha from the SCL, regress portfolio excess returns on the market index.
 - Then, to assess whether the alpha estimate reflects true skill and not just luck, compute the t -statistic of the alpha estimate.
- Even moderate levels of statistical noise make performance evaluation extremely difficult.

Survivorship Bias and Portfolio Evaluation

- Regardless of the performance criterion, some funds will outperform their benchmarks in any year, and some will underperform.
- Performance in one period is not predictive of future performance.
- Limiting a sample of funds to those for which returns are available over an entire sample period introduces **survivorship bias**.

Style Analysis

- *Style analysis* measures the exposures of managed portfolios to asset classes.
- Introduced by William Sharpe.
- Regress fund returns on indexes representing a range of asset classes.
 - Regression coefficient on each index would then measure the fund's implicit allocation to that "style."
 - R^2 of regression would measure percentage of return variability attributable to style choice rather than security selection
 - Intercept measures average return from security selection of the fund portfolio

Table 24.4 Style Analysis for Fidelity's Magellan Fund

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Table 24.4

Style analysis for Fidelity's Magellan Fund

Style Portfolio	Regression Coefficient
T-bill	0
Small cap	0
Medium cap	35
Large cap	61
High P/E (growth)	5
Medium P/E	0
Low P/E (value)	0
Total	100
R-square	97.5

Source: Authors' calculations. Return data for Magellan obtained from finance.yahoo.com/funds and return data for style portfolios obtained from the Web page of Professor Kenneth French: mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Because funds are barred from short position, the regression coefficients are constrained to be either zero or positive and to sum to 100%.

Monthly returns over the five-year period from October 1986 to September 1991.

⁹William F. Sharpe, "Asset Allocation: Management Style and Performance Evaluation," *Journal of Portfolio Management*, Winter 1992, pp. 7–19.

¹⁰Gary Brinson, Brian Singer, and Gilbert Beebower, "Determinants of Portfolio Performance," *Financial Analysts Journal*, May/June 1991.

Source: Authors' calculations. Return data for Magellan obtained from finance.yahoo.com/funds and return data for style portfolios obtained from the Web page of Professor Kenneth French: mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Figure 24.4 Fidelity Magellan Fund Cumulative Return Difference

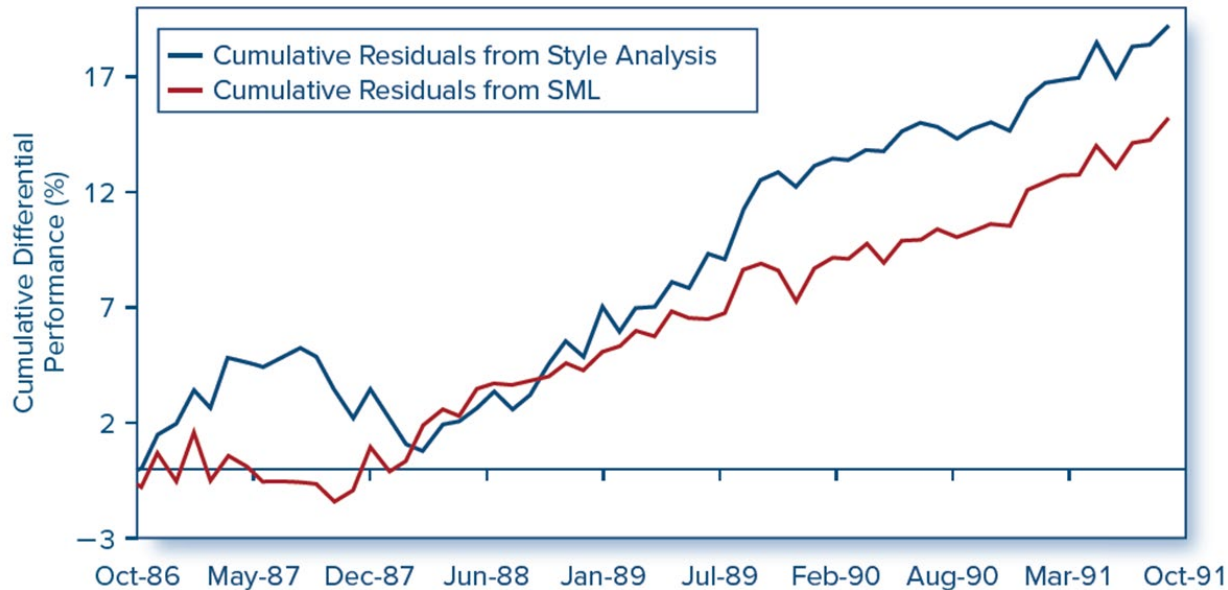


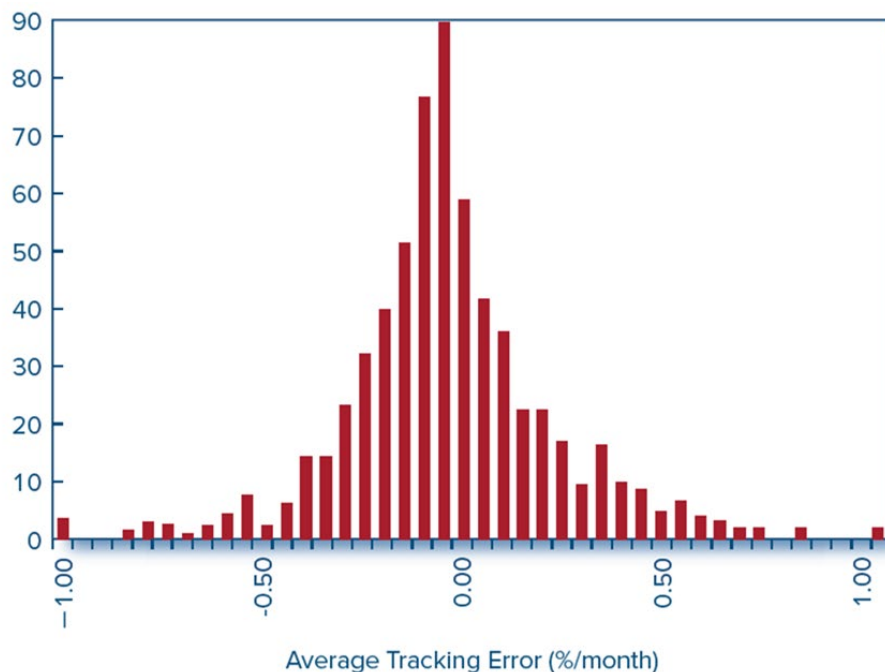
Figure 24.4 Fidelity Magellan Fund cumulative return difference: Fund versus style benchmark and fund versus SML benchmark

Source: Authors' calculations.

Style Analysis

- The fund returns are well represented by three style portfolios.
- Residual variability ($100-97.5=2.5\%$) can be attributed to security selection within asset classes, as well as timing of periodic changes in allocation.
- Importance of the intercept: 32 basis points per month (19.19% over 5 year period)

Figure 24.5 Average Tracking Error for 636 Mutual Funds, 1985 to 1989



Average residual
across 636 mutual
funds

Slightly negative
mean of -0.74% per
month

Figure 24.5 Average tracking error for 636 mutual funds, 1985 to 1989

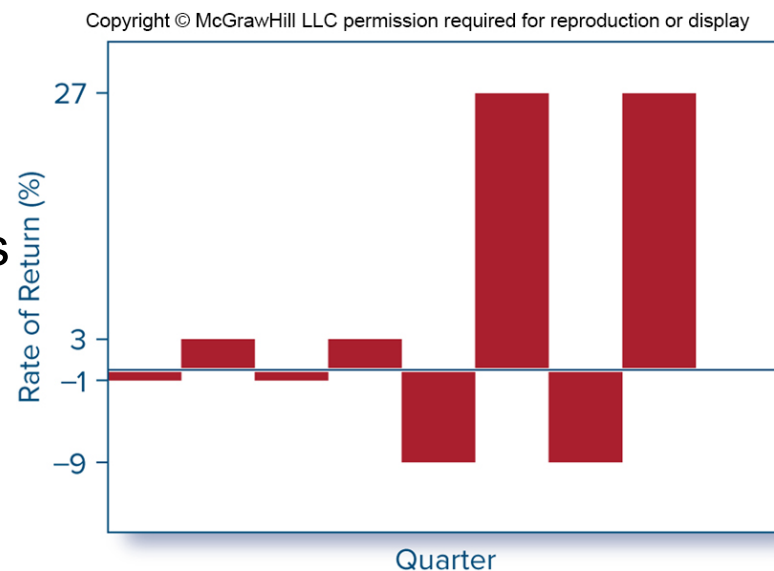
Source: Sharpe, W. F. (1992, Winter). Asset allocation: Management style and performance evaluation. *Journal of Portfolio Management*, 7–19.

Performance Measurement With Changing Portfolio Composition

- We need a very long observation period to measure performance with any precision, even if the return distribution is stable with a constant mean and variance
- Risk-adjustment techniques all assume that portfolio risk is constant over the relevant time period, which isn't necessarily true
- What if the mean and variance are not constant? We need to keep track of portfolio changes

Example 24.3 Changing Portfolio Risk

- Suppose the Sharpe measure of the passive strategy (investing in a market index fund) is 0.4.
- Over an initial period of 52 weeks, a portfolio manager executes a low-risk strategy with an annualized mean excess return of 1% and standard deviation of 2%.
- Over the next 52 weeks, this manager finds that a high-risk strategy is optimal, with an annual mean excess return of 9% and standard deviation of 18%.



Example 24.3 Changing Portfolio Risk

- In initial period, the Sharpe measure is 0.5, which beats the passive strategy.
- For the next four quarters, again the Sharpe measure is 0.5.
- But if we take the eight-quarter sequence as a single measurement period, instead of two independent periods, we get an average excess return of 5% and standard deviation 13.42%, resulting in a Sharpe measure of only 0.37!

Limitations

- Assumptions underlying measures limit their usefulness
- When the portfolio is being actively managed, basic stability requirements are not met (the assumption that portfolio risk, measured by standard deviation or beta, is constant over the relevant time period)
- Practitioners often use benchmark portfolio comparisons to measure performance

Performance Measurement With Changing Portfolio Composition

- Risk-adjustment techniques assume constant portfolio risk, which isn't necessarily true.
- Performance Manipulation and the MRAR.
 - Managers may try to game the system, given their compensation depends on performance.
 - Only measure impossible to manipulate is MRAR.

$$\text{MRAR}(\gamma) = \left[\frac{1}{T} \sum_{t=1}^T \left(\frac{1+r_t}{1+r_{ft}} \right)^{-\gamma} \right]^{\frac{12}{\gamma}} - 1$$

Figure 24.7 MRAR Scores With and Without Manipulation

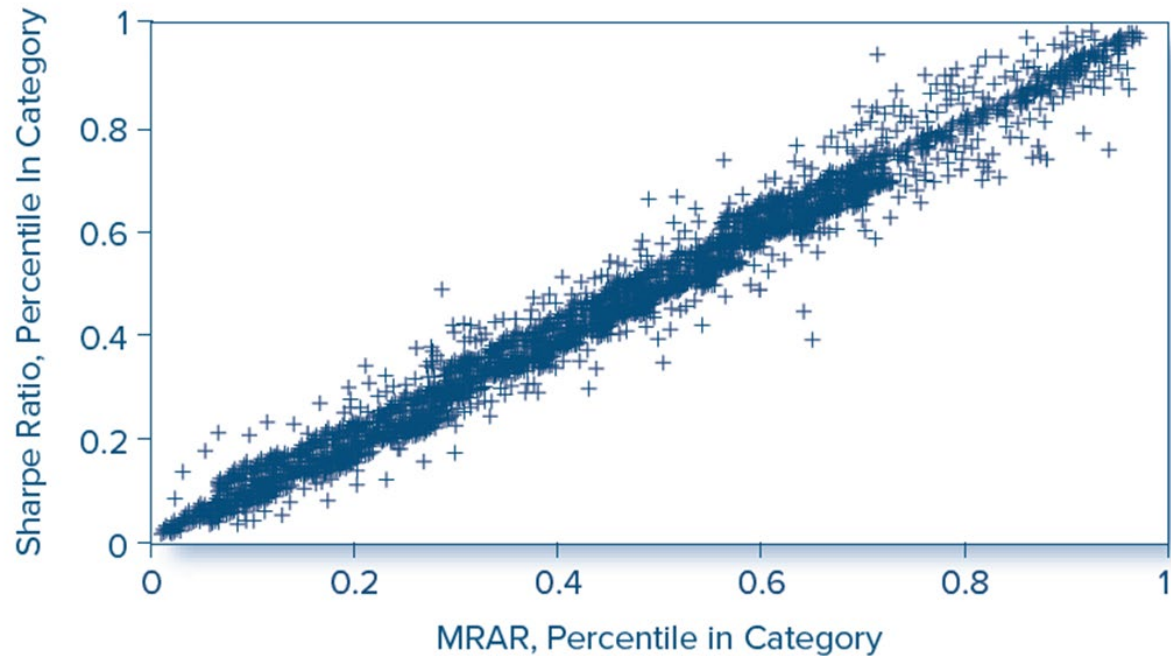


Figure 24.7 Rankings based on Morningstar's RAR versus Sharpe ratio

Source: Sharpe, W. R. (1997). Morningstar performance measures, www.stanford.edu/~wfs Sharpe/art/stars/starsO.htm.

Morningstar rating

- Morningstar computes fund returns as well as a risk measure based primarily on fund performance in its worst years
- The risk-adjusted performance is ranked across funds in a style group and stars are awarded
- Stars are assigned

Percentile	Stars
0–10	1
10–32.5	2
32.5–67.5	3
67.5–90	4
90–100	5

Market Timing

- Suppose we define perfect market timing as the ability to tell (with certainty) at the beginning of each year whether the S&P 500 portfolio will outperform the strategy of rolling over 1-month T-bills through the year.
- Beginning with \$1 on December, 1926 to December 31, 2021.
 - S1: Bills \$21
 - S2: Equities (S&P 500 portfolio) \$10,546
 - S3: Perfect timer \$1,512,355

Table 24.5 Performance of Bills, Equities, and Perfect (Annual) Market Timers

Performance of bills, equities, and perfect (annual) market timers. Initial investment = \$1 (December 31, 1926 to December 31, 2021)

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	Bills	Equities	Perfect timer
Terminal value	\$21	\$10,546	\$1,512,355
Arithmetic average	3.31%	11.96%	16.72%
Standard deviation	3.12%	19.89%	13.34%
Geometric average	3.26%	10.24%	16.16%
Maximum	14.71%	57.35%	57.35%
Minimum*	-0.02%	-44.04%	0.00%
Skew	1.08	-0.45	0.69
Kurtosis	1.09	0.11	-0.14
LPSD	0	12.80%	0

*A negative rate on "bills" was observed in 1940. The Treasury security used in the data series in these early years was actually not a T-bill but a T-bond with 30 days to maturity.

Table 24.5

Performance of bills, equities, and perfect (annual) market timers. Initial investment = \$1

*A negative rate on "bills" was observed in 1940. The Treasury security used in the data series in these early years was actually not a T-bill but a T-bond with 30 days to maturity.

Market Timing & Performance Measurement

- Adjusting portfolio for up and down movements in the market
 - Low Market Return - low β Beta
 - High Market Return - high β Beta
- Shift between stocks and money market instruments or bonds
- Results: higher returns, lower risk (downside is eliminated)
- With perfect ability to forecast behaves like an option

Market Timing

- In its pure form, **market timing** involves shifting funds between a market-index portfolio and a safe asset
- Treynor and Mazuy:

$$r_P - r_f = a + b(r_M - r_f) + c(r_M - r_f)^2 + e_P$$

- Henriksson and Merton:

$$r_P - r_f = a + b(r_M - r_f) + c(r_M - r_f)D + e_P$$

Figure 24.8 Characteristic Lines

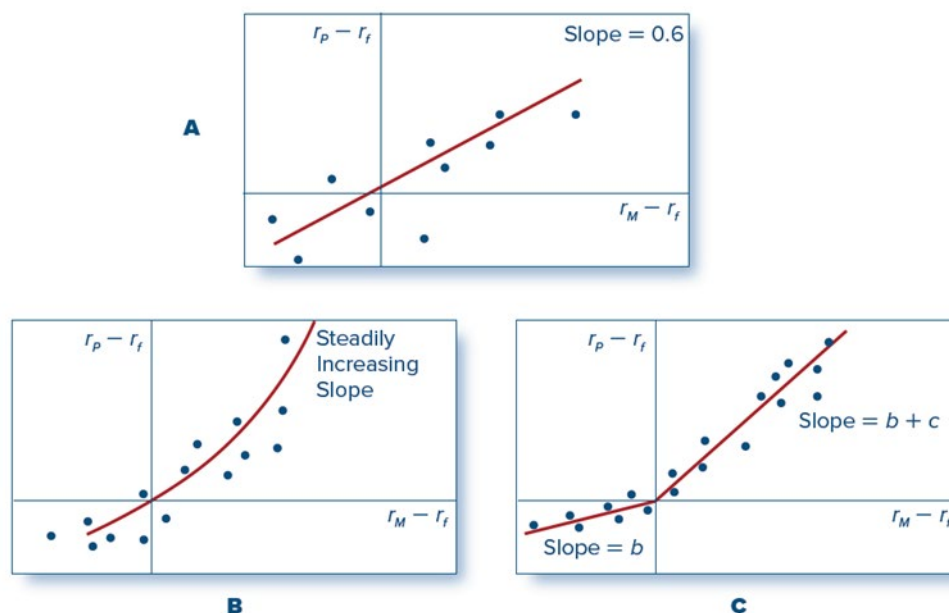


Figure 24.8 Characteristic lines. **Panel A:** No market timing, beta is constant.

Panel B: Market timing, beta increases with expected market excess return.

Panel C: Market timing with only two values of beta.

Characteristic Lines

- A: $r_p - r_f = a + b(r_M - r_f) + e_P$

- B: Treynor and Mazuy

$$r_p - r_f = a + b(r_M - r_f) + c(r_M - r_f)^2 + e_P$$

- C: Henriksson and Merton

$$r_p - r_f = a + b(r_M - r_f) + c(r_M - r_f)D + e_P$$

where D is a dummy variable,

D=1 if $r_M > r_f$ and D=0 otherwise.

Characteristic Lines

- Henriksson and Merton

$$r_p - r_f = a + b(r_M - r_f) + c(r_M - r_f)D + e_P$$

(1) If $r_M > r_f$, we set $D=1$.

$$r_p - r_f = a + b(r_M - r_f) + c(r_M - r_f)(1) + e_P$$

$$r_p - r_f = a + (\underline{b+c})(r_M - r_f) + e_P$$

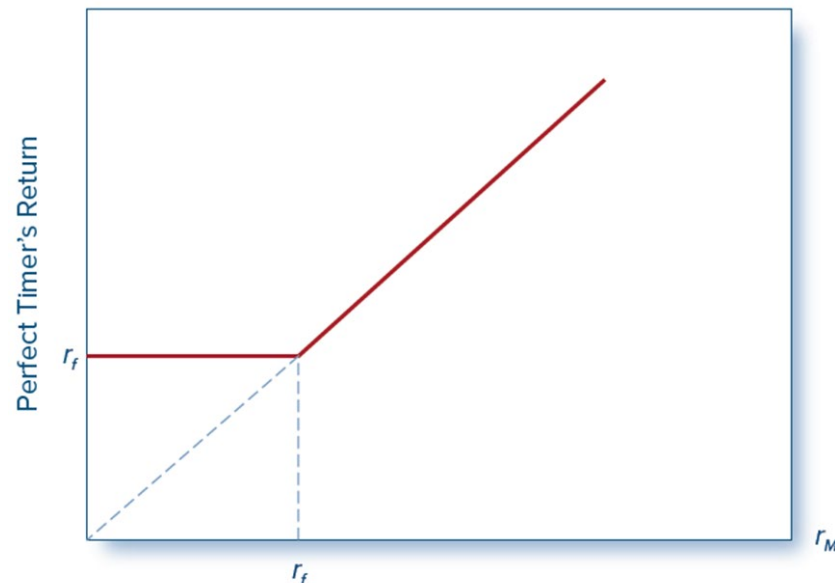
(2) If $r_M \leq r_f$, we set $D=0$.

$$r_p - r_f = a + b(r_M - r_f) + c(r_M - r_f)(0) + e_P$$

$$r_p - r_f = a + \underline{b}(r_M - r_f) + e_P$$

Valuing Market Timing as a Call Option

- Key to valuing market timing ability is to recognize that perfect foresight is equivalent to holding a call option on the equity portfolio—but without having to pay for it!



	$S_T < X$	$S_T \geq X$
Bills	$S_0(1 + r_f)$	$S_0(1 + r_f)$
Call	0	$S_T - X$
Total	$S_0(1 + r_f)$	S_T

$$\begin{aligned}
 \text{MV (Imperfect timer)} &= P \times C \\
 &= (P_1 + P_2 - 1)[2N(0.5 * \sigma_M T^{(1/2)}) - 1]
 \end{aligned}$$

where P is the measure of timing ability ($S_0=1$; $X=\$1 \times e^{rT}$)

Measure of Timing Ability

- P_1 : the proportion of the correct forecasts of bull markets
- P_2 : the proportion of the correct forecasts of bear markets
- $P = P_1 + P_2 - 1$ is the correct measure of timing ability
- Example:
 - Arizona: Rain (37 days) v.s. No Rain (328 days)
 - Always predicts no rain. 90% (=328/365) right. But this should not be the evidence of forecasting ability.
 - $P_1 = (0/37) = \underline{0}$ (Correct proportion of no rain)
 - $P_2 = (328/328) = 1$ (Correct proportion of rain)
 - $P = P_1 + P_2 - 1 = \underline{0} + 1 - 1 = 0$

Performance Attribution Procedures ¹

- Performance attribution studies attempt to decompose overall performance into discrete components that may be identified with a particular level of the portfolio selection process.
- A common attribution system decomposes performance into three components:
 1. Broad asset allocation choices across equity, fixed-income, and money markets.
 2. Industry (sector) choice within each market.
 3. Security choice within each sector.

Performance Attribution Procedures ²

- **Bogey** is designed to measure the returns the portfolio manager would earn if he or she were to follow a completely passive strategy.
 - In this context, “passive” has two attributes.
 1. It means the allocation of funds across broad asset classes is set in accord with a notion of “usual” allocation across sectors.
 2. It means that *within* each asset class, the portfolio manager holds an indexed portfolio.

Asset Allocation Decisions

- Superior performance relative to the bogey is achieved by:
 - Overweighting investments in markets that turn out to perform well.
 - Underweighting investments in poorly performing markets.
- Contribution of asset allocation to superior performance equals the sum over all markets of the excess weight in each market times the return of the index for each market.

Formulas for Attribution

$$r_B = \sum_{i=1}^n w_{Bi} r_{Bi} \text{ \& } r_p = \sum_{i=1}^n w_{pi} r_{pi}$$

$$r_p - r_B = \sum_{i=1}^n w_{pi} r_{pi} - \sum_{i=1}^n w_{Bi} r_{Bi} =$$

$$\sum_{i=1}^n (w_{pi} r_{pi} - w_{Bi} r_{Bi})$$

Where B is the bogey portfolio and p is the managed portfolio

Formulas for Attribution

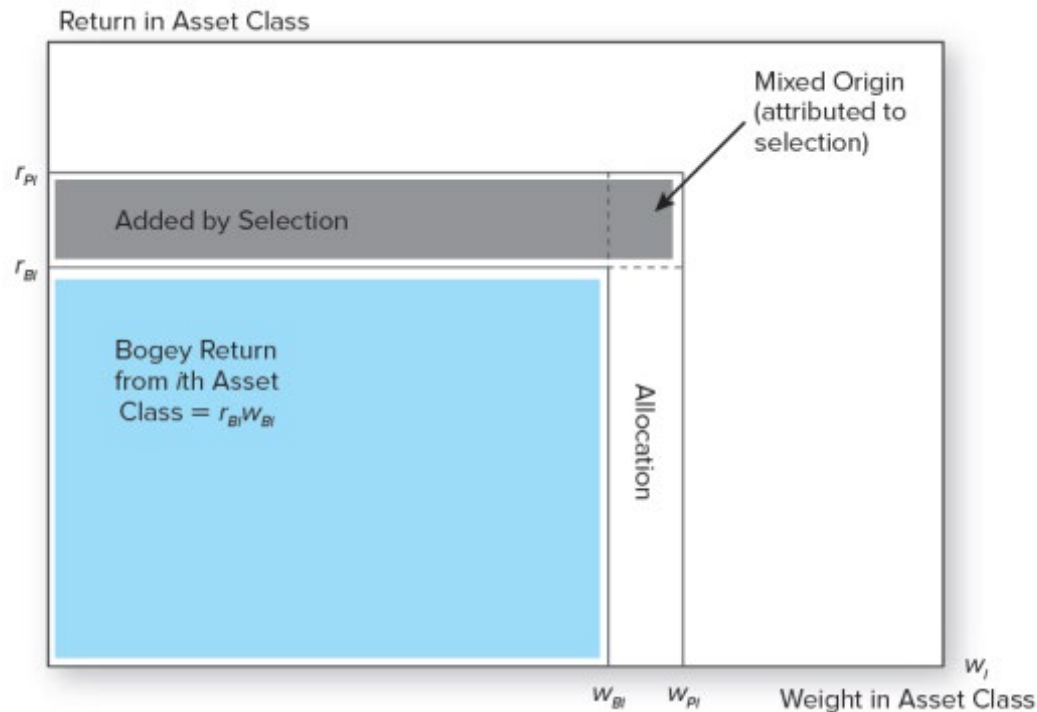


Table 24.6 Performance of the Managed Portfolio

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Table 24.6

Performance of the managed portfolio

Component	Bogey Performance and Excess Return	
	Benchmark Weight	Return of Index during Month (%)
Equity (S&P 500)	0.60	5.81
Bonds (Barclays Aggregate Index)	0.30	1.45
Cash (money market)	0.10	0.48
Bogey = $(0.60 \times 5.81) + (0.30 \times 1.45) + (0.10 \times 0.48) = 3.97\%$		
	Return of managed portfolio	5.34%
	<u>– Return of bogey portfolio</u>	<u>3.97</u>
	Excess return of managed portfolio	1.37%

The managed portfolio is actually invested in the equity, fixed-income, and money markets with weights of 70%, 7%, and 23%, respectively.

Table 24.7 Performance Attribution

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Table 24.7

Performance
attribution

A. Contribution of asset allocation to performance					
	(1)	(2)	(3) = (1) - (2)	(4)	(5) = (3) × (4)
Market	Actual Weight in Market	Benchmark Weight in Market	Active or Excess Weight	Index Return (%)	Contribution to Performance (%)
Equity	0.70	0.60	0.10	5.81	0.5810
Fixed-income	0.07	0.30	-0.23	1.45	-0.3335
Cash	0.23	0.10	0.13	0.48	0.0624
Contribution of asset allocation					0.3099
B. Contribution of selection to total performance					
	(1)	(2)	(3) = (1) - (2)	(4)	(5) = (3) × (4)
Market	Portfolio Performance (%)	Index Performance (%)	Excess Performance (%)	Portfolio Weight	Contribution (%)
Equity	7.28	5.81	1.47	0.70	1.03
Fixed-income	1.89	1.45	0.44	0.07	0.03
Contribution of selection within markets					1.06

Table 24.8 Sector Selection With the Equity Market

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	(1)	(2)	(3)	(4)	(5) = (3) × (4)
	Beginning-of-Month Weights (%)		$=(1)-(2)$		
Sector	Portfolio	S&P 500	Active Weight (%)	Sector Return (%)	Sector Allocation Contribution
Basic materials	1.96	8.3	-6.34	6.9	-0.4375
Business services	7.84	4.1	3.74	7.0	0.2618
Capital goods	1.87	7.8	-5.93	4.1	-0.2431
Consumer cyclical	8.47	12.5	-4.03	8.8	0.3546
Consumer noncyclical	40.37	20.4	19.97	10.0	1.9970
Credit sensitive	24.01	21.8	2.21	5.0	0.1105
Energy	13.53	14.2	-0.67	2.6	-0.0174
Technology	1.95	10.9	-8.95	0.3	-0.0269
Total					1.2898

Table 24.8

Sector selection within the equity market

Table 24.9 Portfolio Attribution: Summary

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Table 24.9

Portfolio attribution:
summary

		Contribution (basis points)
1. Asset allocation		31
2. Selection		
a. Equity excess return (basis points)		
i. Sector allocation	129	
ii. Security selection	<u>18</u>	
	147×0.70 (portfolio weight) =	102.9
b. Fixed-income excess return	44×0.07 (portfolio weight) =	<u>3.1</u>
Total excess return of portfolio		137.0

- Good performance (a positive contribution) derives from overweighting well-performing sectors.
- Good performance also derives from underweighting poorly performing sectors.